

2024 Higher Chemistry Paper 2 - Q7

Section: Chemistry in Society

Topic: Controlling the Rate (Seawater Analysis)

Question summary (Q7):

Seawater contains several dissolved salts.

(a) A sample of seawater had a salinity of 35 parts per thousand.

Using the formula:

$$\text{Salinity} = [\text{Cl}^-] (\text{mg l}^{-1}) \times 0.0018066$$

calculate the chloride ion concentration in g l^{-1} .

(b) Another seawater sample was titrated with standard silver nitrate solution.

(i) (A) Define “standard solution”.

(i) (B) Describe how to prepare and use a pipette to measure the sample.

(ii) Explain why an indicator is used.

Worked Solution:

(a) • $\text{Salinity} = 35 \text{ ppt} = [\text{Cl}^-] \times 0.0018066$

$$\begin{aligned} [\text{Cl}^-] &= 35 \div 0.0018066 = 1.94 \times 10^4 \text{ mg l}^{-1} \\ &= 19.4 \text{ g l}^{-1} \end{aligned}$$

(b) (i) (A) A standard solution is one whose concentration is accurately known.

(B) Pipette preparation:

- Rinse pipette with the solution to be used.
- Use a pipette filler to draw liquid up to the calibration mark.
- Allow liquid to drain into the conical flask, touching the tip to the side to remove the last drop.

(ii) An indicator is used to show the end-point of the titration (visible colour change).

Final Answer:

(a) 19.4 g l^{-1}

(b) (i) (A) Solution of accurately known concentration.

(B) Rinse, fill to mark, deliver with pipette filler.

(b) (ii) To show the end-point by colour change.

Revision Tips:

- $1 \text{ mg l}^{-1} = 0.001 \text{ g l}^{-1}$ when converting units.
- Always rinse pipette with the solution to be measured (not water).
- Indicators in titrations give a clear colour change at the end-point.
- Salinity calculations often use proportional constants to convert between units.